



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE:DATA SCIENCE – 24AD3107 -A.Y. 2026 - 2027

TITLE OF THE PROJECT	Smart Retail Customer Analysis & Prediction	
S. No.	University ID	Name
1	2420030178	S.Niteesh
2	2420030093	Gagan
3	2420030642	P.Nitish
Name of the Guide		Prathusha

Title:

Smart Retail Customer Analysis & Prediction

Abstract:

Retail businesses generate large volumes of raw, unorganised transaction and customer data that is rarely usable in its original form. In this project, raw data collected from retail transactions (customer ID, purchase date, product category, quantity, price, etc.) is first cleaned and converted into structured data through pre-processing steps such as handling missing values, removing duplicates and outliers, correcting data types, and normalising fields. This cleaned data is then organised into structured tables/features (e.g., Recency, Frequency, and Monetary value per customer) using Pandas and NumPy, transforming scattered raw records into an analysis-ready dataset. This structured data is analysed to identify purchasing patterns, product preferences, and spending behaviour, forming the foundation for customer segmentation and prediction.

Using this structured data, K-Means clustering (Scikit-learn) is applied to segment customers into groups such as high-value, frequent, and at-risk customers, while classification/regression models (Logistic Regression, Random Forest, XGBoost) predict future purchase likelihood and customer value. The complete solution is built in Python, using Pandas/NumPy for data structuring, Matplotlib/Seaborn for visualization, Scikit-learn for machine learning, MySQL/SQLite for data storage, and Streamlit/Power BI for an interactive dashboard where retailers can view customer insights, trends, and predictions. By converting raw retail data into structured, analysis-ready information and applying predictive analytics on it, the system aims to improve customer engagement, personalised marketing, and sales planning, resulting in a smarter, more efficient retail management system.

Signature of the Guide

HOD